

* Rank 0 tensor.

$$[T]$$

$$T = \mathbb{T}_0^{(1)}$$

* Rank 1 tensor. * Rank 2 tensor (s).

$$\begin{bmatrix} T_{xx} & 0 & 0 \\ 0 & T_{xx} & 0 \\ 0 & 0 & T_{zz} \end{bmatrix}$$

$$T_{xx} = \mathbb{T}_0^{(1)} - \mathbb{T}_u^{(1)}$$

$$T_{zz} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_u^{(1)}$$

* Rank 2 tensor (a).

$$\begin{bmatrix} 0 & T_{xy} & 0 \\ -T_{xy} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$T_{xy} = \mathbb{M}_z^{(1)}$$

* Rank 3 tensor (ss).

$$\begin{bmatrix} 0 & 0 & T_{xxz} \\ 0 & 0 & -T_{xxz} \\ 0 & 0 & 0 \\ T_{yzx} & -T_{xxz} & 0 \\ T_{xxz} & T_{yzx} & 0 \\ 0 & 0 & T_{yzx} \end{bmatrix}$$

$$T_{xxz} = \mathbb{T}_z^{\beta(1)}$$

$$T_{yzx} = \mathbb{T}_{xyz}^{(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & T_{xxz} \\ 0 & 0 & -T_{xxz} \\ 0 & 0 & 0 \\ T_{yzx} & \frac{T_{xxz}}{2} & 0 \\ -\frac{T_{xxz}}{2} & T_{yzx} & 0 \\ 0 & 0 & -2T_{yzx} \end{bmatrix}$$

$$T_{xxz} = 2\mathbb{M}_{xy}^{(1)}$$

$$T_{yzx} = \mathbb{M}_v^{(1)}$$

$$\frac{T_{xxz}}{2} = \mathbb{M}_{xy}^{(1)}$$

$$2T_{yzx} = 2\mathbb{M}_v^{(1)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} T_{yzx} & T_{yzy} & 0 \\ T_{yzy} & -T_{yzx} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$T_{yzx} = \mathbb{M}_v^{(2)}$$

$$T_{zyy} = \mathbb{M}_{xy}^{(2)}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} T_{xxx} & T_{xyy} & T_{xzz} & 0 & 0 & T_{xxy} \\ T_{xyy} & T_{xxx} & T_{xzz} & 0 & 0 & -T_{xxy} \\ T_{xzz} & T_{xzz} & T_{zzz} & 0 & 0 & 0 \\ 0 & 0 & 0 & T_{yzy} & 0 & 0 \\ 0 & 0 & 0 & 0 & T_{yzy} & 0 \\ T_{xxy} & -T_{xxy} & 0 & 0 & 0 & T_{xyy} \end{bmatrix}$$

$$T_{xxx} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_0^{(2)} - \mathbb{T}_{4u}^{(1)} + 2\mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(1)} - 4\mathbb{T}_u^{(2)}$$

$$T_{xyy} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_{4u}^{(1)} - \mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(1)}$$

$$T_{xzz} = \mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(1)}$$

$$T_{xxy} = \mathbb{T}_{4z}^{\alpha(1)}$$

$$T_{zzz} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_0^{(2)} + 2\mathbb{T}_{4u}^{(1)} + 2\mathbb{T}_4^{(1)} + 4\mathbb{T}_u^{(1)} + 8\mathbb{T}_u^{(2)}$$

$$T_{yzy} = \mathbb{T}_0^{(2)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(2)}$$

$$T_{xyy} = \mathbb{T}_0^{(2)} + 2\mathbb{T}_{4u}^{(1)} - \mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(2)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & T_{xzz} & 0 & 0 & T_{xxy} \\ 0 & 0 & T_{xzz} & 0 & 0 & -T_{xxy} \\ -T_{xzz} & -T_{xzz} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & T_{yzz} & 0 \\ 0 & 0 & 0 & -T_{yzz} & 0 & 0 \\ -T_{xxy} & T_{xxy} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$T_{xzz} = 3\mathbb{T}_u^{(3)}$$

$$T_{xxy} = 2\mathbb{M}_z^{(1)} - 2\mathbb{M}_z^{\alpha(1)}$$

$$T_{yzz} = -\mathbb{M}_z^{(1)} - 4\mathbb{M}_z^{\alpha(1)}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} T_{yzy} & 0 & 0 \\ 0 & T_{yzy} & 0 \\ 0 & 0 & T_{xyy} \end{bmatrix}$$

$$T_{yzy} = \mathbb{T}_0^{(3)} - 2\mathbb{T}_u^{(4)}$$

$$T_{xyy} = \mathbb{T}_0^{(3)} + 4\mathbb{T}_u^{(4)}$$

* Rank 4 tensor (aaa).

$$\begin{bmatrix} 0 & T_{yzz} & 0 \\ -T_{yzz} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$T_{yzz} = -\mathbb{M}_z^{(2)}$$

* Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & 0 & T_{xxy} \\ 0 & 0 & T_{xxy} \\ 0 & 0 & T_{zzz} \\ T_{yzy} & T_{yzz} & 0 \\ T_{yzz} & -T_{yzy} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{aligned}
T_{xxxy} &= \mathbb{M}_z^{(3)} - \mathbb{M}_z^{\alpha(2)} \\
T_{zzxy} &= \mathbb{M}_z^{(3)} + 2\mathbb{M}_z^{(4)} + 2\mathbb{M}_z^{\alpha(2)} \\
T_{yzyz} &= -3\mathbb{T}_u^{(5)} \\
T_{yzzx} &= \mathbb{M}_z^{(4)} - \mathbb{M}_z^{\alpha(2)}
\end{aligned}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} 0 & 0 & 0 & T_{yzyz} & T_{yzzx} & 0 \\ 0 & 0 & 0 & T_{yzzx} & -T_{yzyz} & 0 \\ T_{xyxx} & T_{xyxx} & T_{xyzz} & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{aligned}
T_{yzyz} &= -3\mathbb{T}_u^{(6)} \\
T_{yzzx} &= \mathbb{M}_z^{(6)} - \mathbb{M}_z^{\alpha(3)} \\
T_{xyxx} &= \mathbb{M}_z^{(5)} - \mathbb{M}_z^{\alpha(3)} \\
T_{xyzz} &= \mathbb{M}_z^{(5)} + 2\mathbb{M}_z^{(6)} + 2\mathbb{M}_z^{\alpha(3)}
\end{aligned}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} T_{xxxx} & T_{xxyy} & T_{xxzz} & 0 & 0 & T_{xxxy} & 0 & 0 & T_{xxyx} \\ T_{xxyy} & T_{xxxx} & T_{xxzz} & 0 & 0 & -T_{xxyx} & 0 & 0 & -T_{xxxy} \\ T_{zzxx} & T_{zzxx} & T_{zzzz} & 0 & 0 & T_{zzxy} & 0 & 0 & -T_{zzxy} \\ 0 & 0 & 0 & T_{yzyz} & T_{yzzx} & 0 & T_{yzzz} & T_{yzxz} & 0 \\ 0 & 0 & 0 & -T_{yzzx} & T_{yzzz} & 0 & -T_{yzyz} & T_{yzyz} & 0 \\ T_{xyxx} & -T_{xyxx} & 0 & 0 & 0 & T_{xyxy} & 0 & 0 & T_{xyxy} \end{bmatrix}$$

$$\begin{aligned}
T_{xxxx} &= \mathbb{T}_0^{(1)} + 2\mathbb{T}_0^{(2)} - \mathbb{T}_{4u}^{(1)} + 2\mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(1)} - 4\mathbb{T}_u^{(2)} \\
T_{xxyy} &= \mathbb{T}_0^{(1)} + 2\mathbb{T}_{4u}^{(1)} - \mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(1)} \\
T_{xxzz} &= \mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(1)} + 3\mathbb{T}_u^{(3)} \\
T_{xxxy} &= 2\mathbb{M}_z^{(1)} + \mathbb{M}_z^{(3)} - 2\mathbb{M}_z^{\alpha(1)} - \mathbb{M}_z^{\alpha(2)} + \mathbb{T}_{4z}^{\alpha(1)} \\
T_{xxyx} &= 2\mathbb{M}_z^{(1)} - \mathbb{M}_z^{(3)} - 2\mathbb{M}_z^{\alpha(1)} + \mathbb{M}_z^{\alpha(2)} + \mathbb{T}_{4z}^{\alpha(1)} \\
T_{zzxx} &= \mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(1)} - 3\mathbb{T}_u^{(3)} \\
T_{zzzz} &= \mathbb{T}_0^{(1)} + 2\mathbb{T}_0^{(2)} + 2\mathbb{T}_{4u}^{(1)} + 2\mathbb{T}_4^{(1)} + 4\mathbb{T}_u^{(1)} + 8\mathbb{T}_u^{(2)} \\
T_{zzxy} &= \mathbb{M}_z^{(3)} + 2\mathbb{M}_z^{(4)} + 2\mathbb{M}_z^{\alpha(2)} \\
T_{yzyz} &= \mathbb{T}_0^{(2)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(2)} - 3\mathbb{T}_u^{(5)} \\
T_{yzzx} &= -\mathbb{M}_z^{(1)} + \mathbb{M}_z^{(4)} - 4\mathbb{M}_z^{\alpha(1)} - \mathbb{M}_z^{\alpha(2)} \\
T_{yzzz} &= \mathbb{T}_0^{(2)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(2)} + 3\mathbb{T}_u^{(5)} \\
T_{yzzx} &= -\mathbb{M}_z^{(1)} - \mathbb{M}_z^{(4)} - 4\mathbb{M}_z^{\alpha(1)} + \mathbb{M}_z^{\alpha(2)} \\
T_{xyxx} &= -2\mathbb{M}_z^{(1)} + 2\mathbb{M}_z^{\alpha(1)} + \mathbb{T}_{4z}^{\alpha(1)} \\
T_{xyxy} &= \mathbb{T}_0^{(2)} + 2\mathbb{T}_{4u}^{(1)} - \mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(2)}
\end{aligned}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 & T_{yzyz} & T_{yzzx} & 0 & T_{yzzz} & T_{yzxz} & 0 \\ 0 & 0 & 0 & T_{yzzx} & -T_{yzzz} & 0 & T_{yzzx} & -T_{yzyz} & 0 \\ T_{xyxx} & T_{xyxx} & T_{xyzz} & 0 & 0 & T_{xyxy} & 0 & 0 & -T_{xyxy} \end{bmatrix}$$

$$\begin{aligned}
T_{yzyz} &= \mathbb{T}_0^{(3)} - 2\mathbb{T}_u^{(4)} - 3\mathbb{T}_u^{(6)} \\
T_{yzzx} &= -\mathbb{M}_z^{(2)} + \mathbb{M}_z^{(6)} - \mathbb{M}_z^{\alpha(3)} \\
T_{yzzz} &= -\mathbb{T}_0^{(3)} + 2\mathbb{T}_u^{(4)} - 3\mathbb{T}_u^{(6)} \\
T_{yzxz} &= \mathbb{M}_z^{(2)} + \mathbb{M}_z^{(6)} - \mathbb{M}_z^{\alpha(3)} \\
T_{xyxx} &= \mathbb{M}_z^{(5)} - \mathbb{M}_z^{\alpha(3)} \\
T_{xyzz} &= \mathbb{M}_z^{(5)} + 2\mathbb{M}_z^{(6)} + 2\mathbb{M}_z^{\alpha(3)} \\
T_{xyxy} &= \mathbb{T}_0^{(3)} + 4\mathbb{T}_u^{(4)}
\end{aligned}$$

* Rank 4 tensor (t).

$$\begin{bmatrix}
T_{xxxx} & T_{xxxy} & 0 \\
-T_{xxxxy} & T_{xxxxx} & 0 \\
0 & 0 & T_{zzzz} \\
0 & 0 & T_{yyzz} \\
T_{zzxx} & 0 & 0 \\
T_{xxxy} & T_{xxyy} & 0 \\
0 & T_{yzzz} & 0 \\
0 & 0 & T_{zzxz} \\
T_{xxyy} + T_{yyzz} - T_{yzzz} - T_{zzxz} + T_{zzxx} & -T_{xxxy} & 0 \\
0 & 0 & 0
\end{bmatrix}$$

$$\begin{aligned}
T_{xxxx} &= 3\mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} + 2\mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(1)} - 4\mathbb{T}_u^{(2)} \\
T_{xxxxy} &= \mathbb{T}_{4z}^{\alpha(1)} \\
T_{zzzz} &= 3\mathbb{T}_0^{(1)} + 2\mathbb{T}_{4u}^{(1)} + 2\mathbb{T}_4^{(1)} + 4\mathbb{T}_u^{(1)} + 8\mathbb{T}_u^{(2)} \\
T_{yyzz} &= \mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(1)} + 3\mathbb{T}_u^{(3)} \\
T_{zzxx} &= \mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(1)} - 3\mathbb{T}_u^{(3)} \\
T_{xxyy} &= \mathbb{T}_0^{(1)} + 2\mathbb{T}_{4u}^{(1)} - \mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(1)} \\
T_{yzzz} &= \mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(2)} + 3\mathbb{T}_u^{(5)} \\
T_{zzxz} &= \mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(2)} - 3\mathbb{T}_u^{(5)}
\end{aligned}$$